

THEORY AND METHODS ASYMMETRICAL SINGLE REPLICATE DESIGNS

K.N. Gachii & J.W. Odhiambo

Department of Mathematics, University of Nairobi, P.O. Box 30197, Nairobi,
Kenya.

Key words: Factorial designs; generalized cyclic designs; loss of information.

Summary: Asymmetrical single replicate factorial designs in blocks are constructed using the deletion technique. Results are given that are useful in simplifying expressions for calculating loss of information on main effects and interactions, due to confounding with blocks. Designs for estimating main effects and low order interactions are also given.

1. Introduction

Consider a single replicate factorial experiment involving n factors $F_1, F_2, \dots, F_n$, factor F_i occurring at s_i levels. Let $a = a_1 a_2 \dots a_n$ denote a treatment combination, where a_i ($0 \leq a_i \leq s_i - 1$) is the level of factor F_i . The number of treatment combinations is then given by

$$v = \prod_{i=1}^n s_i \quad (1.1)$$

These v treatment combinations will always be *lexicographically ordered*. That is a treatment combination $a = a_1 a_2 \dots a_n$ appears before another treatment combination $a^* = a_1^* a_2^* \dots a_n^*$ if and only if for the first u such that $a_u \neq a_u^*$ we have $a_u < a_u^*$ for $1 \leq u \leq n$.

Suppose we wish to construct an $s_1 \times s_2 \times \dots \times s_n$ single replicate factorial experiment in blocks. We first construct a single replicate $r_1 \times r_2 \times \dots \times r_n$ preliminary block design, say d_p , using one of the known methods, such that $r_i \geq s_i$, for $i = 1, 2, \dots, n$. We can then select $l_i = r_i - s_i$ levels of the i -th factor of d_p and delete from d_p all treatment combinations where factor F_i occurs at any of the l_i selected levels. The resulting $s_1 \times s_2 \times \dots \times s_n$ single replicate design is referred to as a j -th order deletion design if levels are deleted from j factors.

Bose (1947) laid the foundation of factorial designs. He used finite Euclidean geometry to construct symmetrical factorial designs in blocks. Kishen and Srivastava (1959) extended the method of finite geometries to the construction of balanced confounded asymmetrical factorial designs thereby introducing the idea of deletion.

John and Dean (1975) proposed a simple method of confounding based on the properties of generalized cyclic designs from a set of generating treatments or generators and showed that the confounding patterns could easily be determined from these generators.

More recently Voss (1986) has constructed nearly orthogonal single replicate factorial designs in blocks. He uses the deletion technique where he deletes from the first factor, without loss of generality, to obtain first-order deletion designs. The most recent contribution in this direction is that of Chauhan (1989) who generalized the work by Voss (1986), by constructing efficient single replicate designs using the generalized deletion technique. Starting from an s^n single replicate generalized cyclic design, levels are

deleted from the first m_1 factors, without loss of generality, to obtain an $(s - l)^{m_1} s^{n-m_1}$ deletion design.

The overall objective of the present study is to give results for the general order deletion designs of the form $s^{n-m_1}(s - l)^{m_1}$ which are proper, for $1 \leq l \leq s - 1$ and m_1 less than or equal to the number of generators of the preliminary single replicate generalized cyclic design. The efficient proper single replicate deletion designs of the form $(s - l)s^{n-1}$ given by Chauhan (1989) thus become special cases of the results obtained in this study.

In this paper, the method proposed by John and Dean (1975) is used to construct the preliminary single replicate factorial design, which is always symmetric. That is, factor F_i occurs at $s_i = s$ levels for all $i = 1, 2, \dots, n$. Conditions are given which guarantee the existence of proper or improper deletion designs.

Simple formulas are given for calculating the loss of information, due to confounding with blocks, on main effects and interactions. A simple method of choosing a fraction for estimating main effects and low order interactions is also given.

2. Notation

We shall first assume the fixed effects linear model

$$y_{ah} = \mu + \tau_a + \beta_h + \varepsilon_{ah} \quad (1.2)$$

where y_{ah} denotes the observed yield from treatment combination a in the h -th block; τ_a denotes the effect of the treatment combination a ; β_h denotes

the effect of the h -th block and ε_{ah} are uncorrelated random errors with mean zero and variance σ^2 .

Let $y = (y_{ah})$ and $\tau = (\tau_a)$ denote $v \times 1$ vectors of observations and treatment effects respectively, each lexicographically ordered by a . That is the i -th row corresponds to the i -th treatment combination in the above arrangement of the v treatment combinations.

We shall denote the incidence matrix, the intra block matrix, the diagonal matrix of block sizes and the number of blocks respectively by $\mathbf{N}$, $\mathbf{A}$, $\mathbf{K}$ and $\mathbf{b}$. The i -th row of the incidence matrix $\mathbf{N}$ corresponds to the i -th lexicographically ordered treatment combination $\mathbf{a}$. The $q \times 1$ vectors of ones and of zeros will be denoted by $\mathbf{1}_q$ and $\mathbf{0}_q$ respectively.

A generalized interaction will be denoted by α^x where $x = x_1 x_2 \dots x_n$ such that $x_j = 1$ if F_j is present in the interaction and $x_j = 0$ otherwise. A $v \times 1$ contrast vector will be denoted by c^x where

$$c^x = c^{x_1} \otimes c^{x_2} \otimes \dots \otimes c^{x_n} \quad (1.3)$$

with $c_i^{x_i}$ being an $s_i \times 1$ vector of ones if $x_i = 0$, otherwise it is an $s_i \times 1$ contrast vector. The minimum variance unbiased estimator of the generalized interaction α^x is represented by $c^{x'} \hat{\tau}$, where $\hat{\tau}_i$, the i -th element of $\hat{\tau}$, is the estimate of the fixed effect of the i -th treatment combination.

We shall denote the set of the n factors $\{F_1, F_2, \dots, F_n\}$ by $\{1, 2, \dots, n\}$. Then for a non-empty subset $\{i_1, i_2, \dots, i_r\} \subset \{1, 2, \dots, n\}$, $v(i_1, i_2, \dots, i_r)$ will denote the vector (factorial) space of contrast vectors c^x corresponding to the estimator of the generalized interaction α^x where $x_j = 1$ if $\{j\} \subset \{i_1, i_2, \dots, i_r\}$ and $x_j = 0$ otherwise.

All the notation corresponding to the preliminary designs will carry the subscript p while those corresponding to the deletion designs will carry no subscript.

3. Existence of proper and improper deletion designs

In this section we give some results on the existence of proper and improper deletion designs.

Theorem 1. Consider a deletion design, d , of the form $(s - l_1) \times (s - l_2) \times \dots \times (s - l_n)$ derived from an s^n generalized cyclic design, d_p , with $b_p = s^n/k$ blocks of size $k_p = k$, where l_i ($i = 1, 2, \dots, n$) are the numbers of levels of factor F_i deleted from d_p to obtain d . Then the deletion design d is improper if

$$k[(s - l_1) \times (s - l_2) \times \dots \times (s - l_n)] \neq \gamma s^n$$

where γ is a positive integer.

Proof. Since we have s^n/k blocks, the average number of treatment combinations deleted per block is given by

$$\lambda = k - \frac{k[(s - l_1) \times (s - l_2) \times \dots \times (s - l_n)]}{s^n}$$

Now the deletion design d is improper if unequal numbers of treatment combinations are deleted from different blocks of d_p . Hence d is improper if λ is not a positive integer. Now λ will not be a positive integer only if

$$k[(s - l_1) \times (s - l_2) \times \dots \times (s - l_n)] \neq \gamma s^n$$

for γ a positive integer.

Note that the converse to Theorem 1 is not necessarily true. That is, if γ is a positive integer, it does not necessarily follow that the deletion design d is not improper.

The following result gives deletion designs which can be used for estimation of main effects.

Theorem 2. If s^n is a generalized cyclic design generated by m_1 generators such that $g_1 =$ sum of the first $(n - m + 1)$ rows of an identity matrix of order n and $g_2, g_3, \dots, g_m$ are the last $(m - 1)$ rows of an identity matrix of order n , then there exists a proper deletion design of order m_1 , provided $m_1 \leq m$.

Proof. $\text{HCF}(s, g_i) = 1$ for $i = 1, 2, \dots, m$. Hence $K_i = s$ for $i = 1, 2, \dots, m$ and $k_p = s^m$, $b_p = s^{n-m}$. The general element is given by $u_1g_1 + u_2g_2 + \dots + u_mg_m = u_1(11\dots10\dots0) + u_2(00\dots010\dots0) + \dots + u_m(00\dots01)$ where $u_i = 0, 1, 2, \dots, s - 1$ for $i = 1, 2, \dots, m$. This gives us the initial block. The i -th block is obtained by adding treatment combination $a_1a_2\dots a_{n-m} 0\dots0$ not in the preceding $(j - 1)$ blocks to the initial block, where $a_i = 0, 1, 2, \dots, s - 1$ and $a_1, a_2, \dots, a_{n-m}$ are not all equal. Note that the levels of the m factors $\{F_{n-m+1}, F_{n-m+2}, \dots, F_n\}$ in block j remain the same as in the initial block. Thus to construct an $s^{n-m_1} (s - l)^{m_1}$ deletion design we may delete the same number of treatment combinations from block j , $j = 0, 1, \dots, s^{n-m} - 1$ as from the initial block provided $m_1 \leq m$. Note that $\{F_{n-m_1+1}, F_{n-m_1+2}, \dots, F_n\}$ is a subset of $\{F_{n-m+1}, F_{n-m+2}, \dots, F_n\}$ for $m_1 \leq m$.

4. Some properties of deletion designs

We start by giving some results useful in calculating loss of information due to confounding in blocks. These results were derived by Gachii (1993).

Let $D_{j..}$ be the $r_i \times s_j$ matrix obtained from an $s_j \times s_j$ identity matrix by deleting the t -th row if the t -th level is deleted from factor F_j in the preliminary design d_p to obtain the deletion design d . In our study, $s_j = s$, $j = 1, 2, \dots, n$. We now state the following lemma.

Lemma 1. Suppose levels are deleted in descending order from factor F_j and let P^{a_j} be an $s \times s$ permutation matrix with 1 in the a_j -th column of the 0-th row. Then for $a_j \leq s/2$ we have

$$1'_{(s-l)} D_j P^{a_j} D'_j 1_{(s-l)} = \begin{cases} s, & \text{if } l=0 \\ s-l, & \text{if } a_j=0 \text{ and } l=1, 2, \dots, s-1 \\ s-2l, & \text{if } l=1, 2, \dots, a_j \text{ and } a_j \neq s/2 \\ s-l-a_j, & \text{if } l=a_j+1, a_j+2, \dots, s-a_j-1 \\ 0, & \text{if } l=s-a_j, s-a_j+1, \dots, s-1 \\ & \text{and } a_j \neq 0 \text{ or if } l=a_j=s/2 \end{cases}$$

and for $a_j > s/2$ we have

$$1'_{(s-l)} D_j P^{a_j} D'_j 1_{(s-l)} = \begin{cases} s, & \text{if } l = 0 \\ s - 2l, & \text{if } l = 1, 2, \dots, s - a_j \\ a_j - l, & \text{if } l = s - a_j + 1, \dots, a_j - 1 \\ 0, & \text{if } l = a_j, a_j + 1, \dots, s - 1 \end{cases}$$

Now, let c_j be a contrast vector represented by one of the columns of the matrix $s\mathbf{I}_{(s)} - \mathbf{J}_{(s)}$ which span the space of s -dimensional contrast vectors, where $\mathbf{I}_{(s)}$ is an s -dimensional identity matrix and $\mathbf{J}_{(s)} = \mathbf{1}_{(s)}\mathbf{1}'_{(s)}$, that is an $s \times s$ matrix of 1's. Also let the rows and the columns of the matrix $s\mathbf{I}_{(s)} - \mathbf{J}_{(s)}$ be numbered as $0, 1, 2, \dots, s-1$. Note that if l levels are deleted from factor F_j in the preliminary factorial design to obtain the deletion design, then the contrast vector c_j is one of the columns of the matrix $(s-l)\mathbf{I}_{(s-l)} - \mathbf{J}_{(s-l)}$, where $l = 1, 2, \dots, s-1$.

Lemma 2.

$$c'_j D_j P^{a_j} D'_j c_j = \begin{cases} s(s-l), & \text{if } a_j=0 \text{ and } l=0 \\ -s, & \text{if } l=0 \text{ and } a_j \neq 0 \\ (s-l)(s-l-1), & \text{if } a_j=0 \text{ and } l=1, 2, \dots, s-1 \end{cases}$$

5. Loss of information on low order interactions

Dean (1978) showed that for a given contrast vector c^x , the loss of information φ_x , $0 \leq \varphi_x \leq 1$, due to confounding in blocks, is given by

$$\varphi_x = \frac{c^{x'} \mathbf{N} \mathbf{K}^{-1} \mathbf{N}' c^x}{c^{x'} c^x} \quad (5.1)$$

where $\mathbf{N}$ is the incidence matrix and $\mathbf{K}$ is a diagonal matrix of block sizes.

We shall use the notation

$$d_{a_j} = \mathbf{1}'_{(s-l)} \mathbf{D}_j \mathbf{P}^{a_j} \mathbf{D}_j' \mathbf{1}_{(s-l)}$$

where $\mathbf{1}'_{(s-l)} \mathbf{D}_j \mathbf{P}^{a_j} \mathbf{D}_j' \mathbf{1}_{(s-l)}$ is as given in lemma 1 and

$$d_{a_1 a_2 \dots a_k} = d_{a_1} \times d_{a_2} \times \dots \times d_{a_k}$$

We shall also write

$$d_{a_j}^* = \mathbf{c}_j' \mathbf{D}_j \mathbf{P}^{a_j} \mathbf{D}_j' \mathbf{c}_j$$

where $\mathbf{c}_j$ is as given in section 4.

We consider deletion designs of the form $s^{n-m_1}(s-l)^{m_1}$ with $b = \lambda s^{n-m}$ blocks each of size $k = (1/\lambda)s^{m-m_1}(s-l)^{m_1}$ derived from an s^n generalized cyclic design d_p with

$$k_p = (1/\lambda)s^m \text{ and } b_p = \lambda s^{n-m}$$

where n is the number of factors, m_1 is the order of the deletion design, m is the number of generators and

$$\lambda = \prod_{i=1}^m 1/b_i$$

with $b_i = HCF(s, g_i)$; $i = 1, 2, \dots, m$, as given by John and Dean (1975). The following result on loss of information on two factor interactions is derived by application of (5.1) to these designs.

Lemma 3. The loss of information due to confounding in blocks on any two factor interaction of the form $F_j F_{j'}$ ($j < j'$, $j, j' = 1, 2, \dots, n - m_1$) whose levels were not deleted for both factors in d_p to obtain d , is given by

$$\varphi_x = \frac{\lambda \sum_{a_1} \sum_{a_2} \sum_{a_n} w_{a_1 \dots a_n} d_{a_1 \dots a_{j-1} a_{j+1} \dots a_{j'-1} a_{j'+1} \dots a_n} d_{a_j}^* d_{a_{j'}}^*}{(s-l)^2 (s-l)^{2m_1} s^{n+m-2m_1}}$$

where

$$w_{a_1 a_2, \dots, a_n} = \begin{cases} 1, & \text{if } a = a_1 a_2, \dots, a_n \text{ is in the} \\ & \text{initial block of } d_p \\ 0, & \text{otherwise} \end{cases}$$

6. Confounding in deletion designs

The following results on confounding in generalized cyclic designs are due to John and Dean (1975).

The number of degrees of freedom confounded with blocks for any given interaction, α^x , is given by

$$Y^x = \frac{1}{k} \sum_{a_1} \sum_{a_2} \dots \sum_{a_n} w_{a_1 a_2 \dots a_n} \left(\prod_{j=1}^n z_{a_j}^{x_j} \right) \quad (6.1)$$

where

$$z_{a_j}^{x_j} = \begin{cases} s-1, & \text{if } a_j = 0 \text{ and } x_j = 1 \\ -1 & \text{if } a_j \neq 0 \text{ and } x_j = 1 \\ 1, & \text{if } x_j = 0 \end{cases}$$

and

$$w_{a_1, a_2, \dots, a_n} = \begin{cases} 1, & \text{if } a = a_1 a_2 \dots a_n \text{ is in the initial} \\ & \text{block of } d_p \\ 0, & \text{otherwise} \end{cases}$$

If the number of degrees of freedom in (6.1) is zero the interaction is unconfounded with blocks and if it is $\prod_{j=1}^n (s-1)^{x_j}$ the interaction is totally confounded with blocks.

Consider any interactions between the n factors, say the interactions of the factors $F_{i_1}, F_{i_2}, \dots, F_{i_r}$. Let

$$A = ((a_{ij})) \quad (6.2)$$

where a_{ij} is from the i -th generator $g_i = a_{i1}a_{i2}\dots a_{in}$, $i = 1, 2, \dots, m$ and $j = i_1, i_2, \dots, i_r$.

Consider all the $f \times f$ submatrices contained in the j_1 -th, j_2 -th, ..., j_f -th rows of A and let $h_{j_1, j_2, \dots, j_f}$ be the absolute values of their determinants ($f \leq r$, $f \leq m$). Define H_f as follows

$$H_f = \begin{cases} 1, & \text{if } f = 0 \\ HCF(h_{j_1 j_2 \dots j_f} \setminus \{j_1, j_2, \dots, j_f\} \\ \quad \subset \{i_1, i_2, \dots, i_r\}) & \text{if } 0 < f < m \\ 0, & \text{if } f > m \end{cases} \quad (6.3)$$

In our case the treatment combinations in the initial block are of the form

$$u_1g_1 + u_2g_2 + \dots + u_mg_m \quad (u_i = 0, 1, \dots, s-1; i = 1, 2, \dots, m)$$

The number of treatment combinations in the initial block with i_1 -th, i_2 -th, ..., i_r -th factors all zero is given by $s^{m-r}w_{i_1i_2\dots i_r}$ where

$$w_{i_1i_2\dots i_r} = \begin{cases} \prod_{f=1}^r HCF(s, H_f/H_{f-1}) & \text{if } r \leq m \text{ and } H_r \neq 0 \\ S^{r-g} HCF(w_{j_1\dots j_g} \setminus \{j_1, \dots, j_g\} \\ \quad \subset \{i_1, \dots, i_r\}) & \text{if } g < r \leq m \\ S^{r-m} HCF(W_{j_1\dots j_m} \setminus \{j_1, \dots, j_m\} \\ \quad \subset \{i_1, \dots, i_r\}) & \text{if } r > m \end{cases} \quad (6.4)$$

where g is such that $H_g \neq 0$ and $H_{g+1} = H_{g+2} = \dots = 0$; and $g = 1$ if $H_1 = 0$. Let Y^x be denoted by $Y_{j_1\dots j_h}$ where x has the j_1 -th, j_2 -th, ..., j_h -th digits unity and the remainder zero. Then it can be shown that for the interaction of factors $F_{i_1}, F_{i_2}, \dots, F_{i_r}$ the number of degrees of freedom confounded with blocks is given by

$$Y_{i_1i_2\dots i_r} = w_{i_1\dots i_r} - \sum_{g=1}^{r-1} \left(Y_{j_1j_2\dots j_g} \setminus \{j_1, \dots, j_g\} \subset \{i_1, \dots, i_r\} \right) - 1 \quad (6.5)$$

We now give the following results.

Theorem 3. For the generalized cyclic designs of theorem 2, all the main effects are estimable with full efficiency.

Proof. For the main effect of factor F_j , $j = 1, 2, \dots, n$ we have using (6.3) $H_1 = 1$ and $W_j = 1$ by (6.4).

Thus using (6.5) $Y_j = 0$.

Hence the theorem.

Theorem 4. All the r -factor interactions among any number of the first $(n - m + 1)$ factors, and all the r factor interactions among at least two of the first $(n - m + 1)$ factors, and any number of the last $(m - 1)$ factors, are partially confounded with blocks provided $r \geq m$.

Proof. For the r -factor interaction, $r = 2, 3, \dots, n - m + 1$, we have from equation (6.4)

$$W_{j_1 j_2 \dots j_r} = s^{r-1}$$

and thus equation (6.5) yields

$$0 \neq Y_{j_1 j_2 \dots j_r} < (s - 1)^r$$

Thus all the interactions between at least two of the first $(n - m + 1)$ factors are partially confounded with blocks.

For $r = 3, 4, \dots, n$, the $r \times r$ submatrices of the matrix **A** corresponding to the r -factor interactions between at least two of the first $(n - m + 1)$ factors and any number of the last $(m - 1)$ factors are singular. Therefore using equation (6.4) we obtain

$$w_{j_1 j_2 j_3 \dots j_r} = s^{r-2}$$

and using equation (6.5) we get

$$0 \neq Y_{j_1 j_2 j_3 \dots j_r} < (s - 1)^r$$

We can therefore conclude that all these r -factor interactions are partially confounded with blocks.

Hence the theorem.

Confounding in deletion designs has been studied by Chauhan (1989). Let α^x be a given interaction. Then the factors $F_1, F_2, \dots, F_n$ or simply $\{1, 2, \dots, n\}$ can be partitioned into three mutually exclusive and exhaustive sets Ω_1, Ω_2 and Ω_3 as follows: Ω_1 contains the factors whose levels were not deleted from d_p to obtain d , that is $\{1, 2, \dots, n - m_1\}$; Ω_2 contains the factors whose levels were deleted from d_p to obtain d and these factors are not in the factorial space V_x , that is the factors $\{n - m_1 + 1; n - m_1 + 2, \dots, n - m_1 + a\}$; Ω_3 contains the factors whose levels were deleted from d_p to obtain d and these factors are in the factorial space V_x , that is factors $\{n - m_1 + a + 1, n - m_1 + a + 2, \dots, n\}$; $a = 0, 1, 2, \dots, m_1$.

We shall write the factorial space V_x as $V(j_1, j_2, \dots, j_r)$ if $x_{j_1} = x_{j_2} = \dots = x_{j_r} = 1$ and all other x_j 's are zero, where $\{j_1, j_2, \dots, j_r\} \subseteq \{1, 2, \dots, n\}$. Let $\{i_1, i_2, \dots, i_r\} \subset \{1, 2, \dots, n - m_1\}$, then Chauhan (1989) showed that if the contrast vector $c^x \in V(i_1, i_2, \dots, i_r, n - m_1 + a + 1, n - m_1 + a + 2, \dots, n)$ and if

$$c_p^x = \mathbf{D}' c^x \quad (6.6)$$

then $c_p^x \in \oplus V_p(i_1, i_2, \dots, i_r, g, n - m_1 + a + 1, \dots, n)$ where $g \in \mathbb{P}(\Omega_2)$; that is g belongs to the power set of Ω_2 and $\oplus$ denotes the direct sum. Where

$$\mathbf{D} = \mathbf{D}_1 \otimes \mathbf{D}_2 \otimes \dots \otimes \mathbf{D}_n$$

again where D_j is as defined in lemma 1

The following result follows by application of theorems 3, 4 and equation (6.6):

For the deletion designs of order m_1 derived from generalized cyclic designs of theorem 2, all the main effects are fully estimable provided $m_1 < m$.

Loss of information

Using lemmas 1, 2 and 3, we have the following two results on loss of information

- (a) The loss of information on two-factor interactions of the form $F_j F_{j'}$, $j < j' = 1, 2, \dots, n - m + 1$ for the generalized cyclic designs of theorem 2 is given by

$$\varphi_x = \frac{1}{s-1} \quad (6.7)$$

- (b) If $m = 2$, then for the deletion designs of the form $s^{n-1}(s-l)$ the loss of information on two-factor interactions of the form $F_j F_{j'}$, $j < j'$, $j, j' = 1, 2, \dots, n-1$ is given by

$$\varphi = \frac{s^3 - sl(2s-l)}{s(s-1)(s-l)^2} \quad (6.8)$$

provided that $l < s/2$.

Example. Suppose we have 3 different types of chicken feeds F_1, F_2, F_3 and we have reason enough to investigate the effect of these feeds on the weight gain of the chickens at six equally spaced levels of each feed, for example 0,

10, ..., 50 grams. To investigate which feed and which level or combination of levels of the three feeds is most effective, we can construct a 6^3 generalized cyclic design generated by $g_1 = 1 \ 1 \ 0$ and $g_2 = 0 \ 0 \ 1$ (using theorem 2). The levels of feed are denoted by 0, 1, ..., 5 respectively.

This design will have $b_p = 6$ homogeneous groups (blocks) of chickens each of size $k_p = 6^2$ chickens. All effects are fully estimable except the two-factor interaction $F_1 F_2$ which is estimable with a loss of information

$$\varphi_x = \frac{1}{6-1} = 0.2$$

using (6.7).

Note that if we reduce the block sizes by not giving the 50 grams (level 5) of F_3 , we get a $6^2 \times 5$ deletion design in $b = 6$ blocks, each of size $k = 6 \times 5 = 30$ chickens. Again all effects are fully estimable except the two-factor interaction $F_1 F_2$ which is estimable with a loss of information

$$\varphi_x = \frac{6^3 - 6(12-1)}{6 \times 5^3} = 0.2 \quad (l=1)$$

using (6.8).

If we reduce the block sizes by not giving the 40 and 50 grams (levels 4 and 5) of F_3 we get a $6^2 \times 4$ deletion design in $b = 6$ blocks each of size $k = 6 \times 4 = 24$ chickens. All the interactions are estimable except the two-factor interaction $F_1 F_2$ which is estimable with a loss of information

$$\varphi_x = \frac{6^3 - 6 \times 2(12-2)}{6 \times 5 \times 4^2} = 0.2 \quad (l=2)$$

using (6.8).

7. Concluding remarks

Deletion designs give us a simpler method of choosing a fraction for estimating main effects and low order interactions from a preliminary symmetric design. We note that the confounding patterns are easily determined for the fraction so chosen. Expressions for loss of information in terms of the number of levels of the factors in the preliminary design and the number of levels deleted from the j factors, have been derived.

The efficiency of these deletion designs with respect to loss of information compares favourably with the generalized cyclic designs. For the example above loss of information on the confounded interaction $F_1 F_2$ is the same in all three designs, but $6^2(6 - 2)$ deletion design requires a smaller number of treatment combinations. Hence it is economical.

8. References

- BOSE, R.C. (1947). Mathematical theory of symmetrical factorial design. *Sankhyā*, **8**, 107–166.
- CHAUHAN, C.K. (1989). Construction of efficient single replicate designs. *Preprint*.
- DEAN, A.M. (1978). The analysis of interactions in single replicate generalized cyclic designs. *Journal of the Royal Statistical Society, Ser. B*, **40**, 70–89.
- GACHII, K.M. (1993). On the construction of deletion designs. *Ph.D. thesis, University of Nairobi*.
- JOHN, J.A. & DEAN, A.M. (1975). Single replicate factorial experiments in generalized cyclic designs: I Symmetrical arrangements. *Journal of the Royal Statistical Society, Ser. B*, **37**, 63–71.

- VOSS, D.T. (1986). First order deletion designs and the construction of efficient nearly orthogonal factorial designs in small blocks. *Journal of the American Statistical Association*. **81**, 813–818.